

MATA37H3 — Week 02

Answer Key & Misconception Notes

EXAMPLE 1 — Answer, step by step

Step 1. Antidifferentiate $3x^8 - 18x^5 + 1$ term by term using the power rule $\int x^n dx = \frac{x^{n+1}}{n+1}$:
 $\int 3x^8 dx = \frac{3x^9}{9} = \frac{x^9}{3}$; $\int -18x^5 dx = -\frac{18x^6}{6} = -3x^6$; $\int 1 dx = x$.

Step 2. Combine with a single constant: $\int (3x^8 - 18x^5 + 1) dx = \frac{x^9}{3} - 3x^6 + x + C$.

Step 3. For $\int \sqrt[3]{x} dx$, rewrite as $\int x^{1/3} dx$; add 1 to the exponent: $\frac{1}{3} + 1 = \frac{4}{3}$.

Step 4. Divide by the new exponent, i.e. multiply by its reciprocal $\frac{3}{4}$: $\int x^{1/3} dx = \frac{3}{4}x^{4/3} + C$.

Common misconception: Using the power rule on $x^{1/3}$ but writing the new exponent as $1/3 - 1 = -2/3$ without flipping the sign correctly, or forgetting to invert-and-multiply by the reciprocal of the new exponent ($\frac{3}{4}$, not $\frac{4}{3}$ inverted incorrectly).

EXAMPLE 2 — Answer, step by step

Step 1. Recall from the derivative table that $\frac{d}{dx}[\sec x] = \sec x \tan x$; reversing this gives
 $\int \sec x \tan x dx = \sec x + C$.

Step 2. Recall $\frac{d}{dx}[\ln|x+5|] = \frac{1}{x+5} \cdot 1 = \frac{1}{x+5}$ (chain rule, derivative of the inside $x+5$ is 1); reversing gives $\int \frac{dx}{x+5} = \ln|x+5| + C$.

Common misconception: Confusing $\sec x \tan x$'s antiderivative with $\tan x$'s (writing $\tan x + C$ instead of $\sec x + C$) — these are two different “guess and check” facts that get mixed up.

EXAMPLE 3 — Answer, step by step

Step 1. Domain: $F(x) = \ln|3x|$ requires $3x \neq 0$; restricting to $x > 0$ (as the problem asks), $3x > 0$ so $|3x| = 3x$ and $F(x) = \ln(3x)$.

Step 2. Differentiate with the chain rule: $F'(x) = \frac{1}{3x} \cdot \frac{d}{dx}[3x] = \frac{1}{3x} \cdot 3 = \frac{1}{x}$ — the factor of 3 from the inside function cancels the 3 in the denominator.

Step 3. Since $F'(x) = \frac{1}{x}$ on $x > 0$, F is indeed a valid antiderivative of $\frac{1}{x}$ there.

Step 4. Identify C : by the log law $\ln(3x) = \ln 3 + \ln x$, so $F(x) = \ln x + \ln 3$ — matching the general form $\ln x + C$ with $C = \ln 3$.

Common misconception: Computing $\frac{d}{dx} \ln(3x)$ as $\frac{1}{3}$ (forgetting the chain rule multiplies by the derivative of the *inside*, which is 3, and the 3s cancel to leave exactly $1/x$, not $1/(3x)$ or $1/3$).

EXAMPLE 4 — Answer, step by step

Step 1. The rule to check in each case: the dummy (integration) variable — the one attached to the differential, e.g. dt — must never be the same letter as the free/outer variable x ; but x is completely allowed to appear elsewhere (as the upper/lower limit, or as a plain parameter inside the integrand), since none of that conflicts with t being the variable of integration.

Step 2. (a) $F(x) = \int_0^x \frac{t}{1+t^8} dt$: dummy variable is t , x only appears as the upper limit. **Valid.**

Step 3. (b) $F(x) = \int_0^x \frac{x}{1+t^8} dt$: the differential is still dt , so t is the dummy variable; x appears as the upper limit *and* as a constant-with-respect-to- t factor in the integrand, which is perfectly fine — it simply means $F(x) = x \int_0^x \frac{dt}{1+t^8}$, a well-defined function of x . **Valid**

(this corrects an earlier, over-cautious version of this key, which is why the misconception below is worth reading carefully).

Step 4. (c) $F(x) = \int_3^0 \frac{t}{1+x^2+t^8} dt$: the differential is dt , so t is again the dummy variable; the limits are fixed numbers (3 and 0, not functions of x), and x appears only as a parameter (x^2) inside the integrand. A “backwards” pair of constant limits is not a problem — it only contributes an overall minus sign, $\int_3^0 (\dots) dt = -\int_0^3 (\dots) dt$. **Valid.**

Common misconception: Assuming that *any* appearance of the outer variable x inside the integrand invalidates the definition. It doesn't — the only actual violation would be x itself acting as the dummy variable, e.g. something like $\int_0^x \frac{x}{1+x^8} dx$, where the differential dx collides with x also being the limit. None of (a), (b), (c) do that, so all three are valid.

EXAMPLE 5 — Answer, step by step

Step 1. The upper limit is bare x (not a function of x), so FTC Part 1 applies directly with $f(t) = \cos(t^2)$: $F'(x) = f(x) = \cos(x^2)$.

Common misconception: Adding an extra chain-rule factor when the upper limit is literally x — no extra multiplication is needed here, unlike Problems 6–7 where the limit is a genuine function of x .

EXAMPLE 6 — Answer, step by step

Step 1. The variable is in the *lower* limit, so first flip the integral to put it on top: $F(x) = \int_x^7 \sin^3(\sqrt{t}) dt = -\int_7^x \sin^3(\sqrt{t}) dt$.

Step 2. Now the upper limit is bare x , so FTC Part 1 applies with $f(t) = \sin^3(\sqrt{t})$: $\frac{d}{dx} \left[-\int_7^x f(t) dt \right] = -f(x) = -\sin^3(\sqrt{x})$.

Common misconception: Forgetting to flip the sign when the variable is in the *lower* limit, and reporting $F'(x) = \sin^3(\sqrt{x})$ without the minus sign.

EXAMPLE 7 — Answer, step by step

Step 1. Both limits depend on x , so split at a convenient constant, $t = 0$, with $f(t) = \frac{t}{1+t^8}$:

$$F(x) = -\int_0^{\sin x} f(t) dt + \int_0^{e^x} f(t) dt.$$

Step 2. Differentiate the upper-limit piece with $g(x) = e^x$, $g'(x) = e^x$: $\frac{d}{dx} \left[\int_0^{e^x} f dt \right] = f(e^x) \cdot e^x = \frac{e^x}{1 + e^{8x}} \cdot e^x$.

Step 3. Differentiate the lower-limit piece with $h(x) = \sin x$, $h'(x) = \cos x$: $\frac{d}{dx} \left[- \int_0^{\sin x} f dt \right] = -f(\sin x) \cdot \cos x = -\frac{\sin x}{1 + \sin^8 x} \cdot \cos x$.

Step 4. Combine: $F'(x) = \frac{e^x}{1 + e^{8x}} \cdot e^x - \frac{\sin x}{1 + \sin^8 x} \cdot \cos x$.

Common misconception: Only differentiating with respect to the upper limit and forgetting the lower limit is *also* a function of x here — both limits need their own chain-rule term, with a minus sign on the lower-limit term.

EXAMPLE 8 — Answer, step by step

Step 1. (a) Test with a counterexample: let $G(x) = \int_0^x f(t)dt + 5$. Then $G'(x) = f(x)$ (so G genuinely is an antiderivative of f), but $G(x) \neq \int_0^x f(t)dt$ unless the extra 5 happens to be 0. So the claim “ $G(x) = \int_0^x f(t)dt$ must hold” is **False**.

Step 2. (b) This is exactly the “ $+C$ ” fact from Page 1: both $G(x)$ and $\int_0^x f(t)dt$ are antiderivatives of f on the single interval $\mathbb{R}$, so they differ by a constant. **True**.

Common misconception: Believing that *every* antiderivative must be expressible with lower limit exactly 0 — the constant C can equally well be realized using $\int_{x_0}^x f dt$ for any other base point x_0 ; there is no canonical choice.

EXAMPLE 9 — Answer, step by step

Step 1. Find an antiderivative term by term: $\int e^x dx = e^x$; $\int e^{-x} dx = -e^{-x}$ (chain rule on the exponent $-x$); $\int -\cos(2x) dx = -\frac{1}{2} \sin(2x)$ (check: $\frac{d}{dx} [-\frac{1}{2} \sin 2x] = -\cos 2x$ ✓). So $F(x) = e^x - e^{-x} - \frac{1}{2} \sin(2x)$.

Step 2. Apply FTC Part 2: $\int_0^1 (e^x + e^{-x} - \cos 2x) dx = F(1) - F(0)$.

Step 3. $F(1) = e - e^{-1} - \frac{1}{2} \sin 2$; $F(0) = 1 - 1 - \frac{1}{2} \sin 0 = 0$.

Step 4. Subtract: $F(1) - F(0) = e - e^{-1} - \frac{1}{2} \sin 2$.

Common misconception: Mis-antidifferentiating e^{-x} as e^{-x} instead of $-e^{-x}$ (a sign error from the chain rule on the exponent $-x$), which flips the sign of that whole term in the final answer.

EXAMPLE 10 — Answer, step by step

Step 1. Use the row $\sec^2 x \rightarrow \tan x$ from the derivative table (never $\sec x \tan x$, which is the derivative *of* $\sec x$, not its antiderivative): $\int \sec^2 x dx = \tan x + C$.

Step 2. Evaluate: $[\tan x]_{\pi/4}^{\pi/3} = \tan(\pi/3) - \tan(\pi/4) = \sqrt{3} - 1$.

Common misconception: Writing the antiderivative of $\sec^2 x$ as $\sec x \tan x$ (confusing it with the derivative *of* $\sec x$) instead of $\tan x$.

EXAMPLE 11 — Answer, step by step

Step 1. Find where the curves meet: $x^2 + 3 = 3x + 1 \Rightarrow x^2 - 3x + 2 = 0 \Rightarrow (x-1)(x-2) = 0 \Rightarrow x = 1, 2$.

Step 2. Decide which curve is on top on $[1, 2]$ by testing $x = 1.5$: line gives $3(1.5) + 1 = 5.5$; parabola gives $1.5^2 + 3 = 5.25$. The line is larger, so the line is on top.

Step 3. Set up the area integral, top minus bottom: $\text{Area} = \int_1^2 [(3x + 1) - (x^2 + 3)] dx = \int_1^2 (-x^2 + 3x - 2) dx$.

Step 4. Antidifferentiate and evaluate: $\left[-\frac{x^3}{3} + \frac{3x^2}{2} - 2x\right]_1^2$. At $x = 2$: $-\frac{8}{3} + 6 - 4 = -\frac{2}{3}$. At $x = 1$: $-\frac{1}{3} + \frac{3}{2} - 2 = -\frac{5}{6}$.

Step 5. Subtract: $-\frac{2}{3} - \left(-\frac{5}{6}\right) = -\frac{4}{6} + \frac{5}{6} = \frac{1}{6}$.

Common misconception: Integrating $(x^2 + 3) - (3x + 1)$ (top minus bottom reversed) without first checking which curve is actually larger on $[1, 2]$, producing a negative “area.”

EXAMPLE 12 — Answer, step by step

Step 1. Average value formula: $\text{avg} = \frac{1}{b-a} \int_a^b f(x) dx = \frac{1}{3-0} \int_0^3 (9-x^2) dx$.

Step 2. Antidifferentiate and evaluate: $\frac{1}{3} \left[9x - \frac{x^3}{3} \right]_0^3 = \frac{1}{3} [(27-9) - 0] = \frac{1}{3}(18) = 6$.

Step 3. Solve $f(c) = 6$: $9 - c^2 = 6 \Rightarrow c^2 = 3 \Rightarrow c = \pm\sqrt{3}$; only $c = \sqrt{3}$ lies in $[0, 3]$.

Common misconception: Forgetting to divide by $b-a$ when computing the average value — reporting the bare integral 18 as if it were the average, instead of $18/3 = 6$.

EXAMPLE 13 — Answer, step by step

Step 1. Antidifferentiate term by term: $\int 4x^3 dx = x^4$; $\int -3x^2 dx = -x^3$, giving $F(x) = x^4 - x^3$.

Step 2. Evaluate at the endpoints: $F(1) = 1^4 - 1^3 = 1 - 1 = 0$; $F(-1) = (-1)^4 - (-1)^3 = 1 - (-1) = 2$.

Step 3. Apply FTC Part 2: $\int_{-1}^1 (4x^3 - 3x^2) dx = F(1) - F(-1) = 0 - 2 = -2$.

Common misconception: Treating $(-1)^3$ as $+1$ instead of -1 when evaluating $F(-1)$ — an odd-power sign error that silently turns a subtraction into an addition.

EXAMPLE 14 — Answer, step by step

Step 1. The upper limit is $g(x) = \ln x$, a genuine function of x (not bare x), so use the chain-rule extension of FTC Part 1 with $f(t) = t^2$: $F'(x) = f(g(x)) \cdot g'(x)$.

Step 2. Compute the two pieces: $f(g(x)) = f(\ln x) = (\ln x)^2$; $g'(x) = \frac{d}{dx}[\ln x] = \frac{1}{x}$.

Step 3. Multiply: $F'(x) = (\ln x)^2 \cdot \frac{1}{x} = \frac{(\ln x)^2}{x}$.

Common misconception: Treating $\ln x$ in the upper limit like the “bare x ” case (Problem 5) and skipping the extra factor of $g'(x) = 1/x$ that the chain rule requires whenever the limit is a function of x rather than x itself.

EXAMPLE 15 — Answer, step by step

Step 1. Average value: $\text{avg} = \frac{1}{\pi/4 - 0} \int_0^{\pi/4} \sec^2 x \, dx = \frac{4}{\pi} [\tan x]_0^{\pi/4} = \frac{4}{\pi}(1 - 0) = \frac{4}{\pi}$.

Step 2. Set up the MVT equation: solve $\sec^2 c = \frac{4}{\pi}$ for $c \in [0, \frac{\pi}{4}]$.

Step 3. Rewrite in cosine: $\sec^2 c = \frac{4}{\pi} \iff \cos^2 c = \frac{\pi}{4} \iff \cos c = \frac{\sqrt{\pi}}{2}$ (taking the positive root, since $c \in [0, \frac{\pi}{4}]$ means $\cos c > 0$).

Step 4. So $c = \arccos\left(\frac{\sqrt{\pi}}{2}\right) \approx \arccos(0.886) \approx 0.482$ radians — and this does lie inside $[0, \frac{\pi}{4}] \approx [0, 0.785]$, confirming the Mean Value Theorem for Integrals.

Common misconception: Solving $\sec^2 c = 4/\pi$ by taking $c = \text{arcsec}(\sqrt{4/\pi})$ directly without first converting to $\cos c$, which is much easier to invert exactly here — or forgetting to check that the resulting c actually lies inside the given interval.